

PERFORMAX EPMS

Enterprise Performance Management System — Technical Architecture & System Documentation

Version: 1.0.0 **Status:** Production-Ready **Date:** September 2026

Automated Tests: 25 Passed (100%)

1. Database Schema Architecture

The system is backed by Django ORM with native PostgreSQL (Production) and SQLite (Development) support. Tables enforce foreign key cascades, unique constraints, and strict typing.

1.1 Authentication & Security (apps.accounts)

- User (accounts_user):** id (BigAuto PK), username (CharField unique), email (EmailField unique), role (SUPER_ADMIN, HR, MANAGER, EMPLOYEE, INTERN), password_change_required (Bool), is_active , is_staff , is_superuser , date_joined , last_login .
- EmailOTP (accounts_emailotp):** id (UUID PK), email (CharField), otp_code (CharField), expires_at (DateTime), is_used (Bool).
- PasswordResetToken (accounts_passwordresettoken):** id (UUID PK), user_id (FK User), token_hash (CharField), expires_at (DateTime), is_used (Bool).
- RolePermission (accounts_rolepermission):** id (UUID PK), role (CharField), permission_code (CharField), description (CharField). Persists dynamic RBAC mappings.

1.2 Organization Governance (apps.organization)

- Department (organization_department):** id (UUID PK), code (CharField unique), name (CharField), description (TextField), is_active (Bool).
- JobLevel (organization_joblevel):** id (BigAuto PK), level_code (CharField unique), level_name (CharField), level_rank (Int).
- Position (organization_position):** id (BigAuto PK), position_code (CharField unique), position_name (CharField), level_id (FK JobLevel).
- Role (organization_role):** id (BigAuto PK), role_name (CharField unique), description (TextField).
- Team (organization_team):** id (UUID PK), name (CharField), department_id (FK Department), manager_id (FK User), description (TextField).
- TeamMembership (organization_teammembership):** id (UUID PK), team_id (FK Team), employee_id (FK EmployeeProfile), is_lead (Bool), joined_at (DateTime).

- **FinancialYear (organization_financialyear):** `id` (BigAuto PK), `title` (CharField), `start_date` , `end_date` (DateField), `is_current` (Bool), `status` (CharField).
- **PerformanceCategory (organization_performancecategory):** `id` (BigAuto PK), `name` (CharField), `min_score` , `max_score` , `rating_value` (Int), `grade` (CharField).

1.3 Employee Information (apps.employees)

- **EmployeeProfile (employees_employeeprofile):** `id` (UUID PK), `user_id` (OneToOne User), `employee_id` (CharField unique, e.g. EMP-1001, INT-2026-001), `department_id` (FK Department), `position_id` (FK Position), `manager_id` (FK User), `joining_date` , `phone_number` , `birth_place` , `competencies` (JSON).

1.4 Performance & Appraisals (apps.performance)

- **PerformanceCycle (performance_performancecycle):** `id` (UUID PK), `title` (CharField), `cycle_type` (CharField), `start_date` , `end_date` (DateField), `status` (DRAFT, ACTIVE, IN_REVIEW, CALIBRATION, CLOSED).
- **EvaluationCriterion (performance_evaluationcriterion):** `id` (UUID PK), `cycle_id` (FK PerformanceCycle), `name` (CharField), `maximum_score` , `weight` (Decimal).
- **Appraisal (performance_appraisal):** `id` (UUID PK), `employee_id` (FK EmployeeProfile), `cycle_id` (FK PerformanceCycle), `reviewer_id` (FK User), `appraisal_type` (CharField), `status` (DRAFT, SELF_SUBMITTED, MANAGER_REVIEWED, CALIBRATED, PUBLISHED), `overall_score` (Decimal), `classification` (CharField), `allow_intern_reply` (Bool).
- **AppraisalRating (performance_appraisalrating):** `id` (UUID PK), `appraisal_id` (FK Appraisal), `criterion_id` (FK EvaluationCriterion), `score` (Decimal), `comments` (TextField).
- **TechnicalCapabilityParameter (performance_technicalcapabilityparameter):** `id` (UUID PK), `cycle_id` (FK PerformanceCycle), `name` (CharField), `category` (CharField), `benchmark_score` , `weight` (Decimal), `created_by_id` (FK User).
- **TechnicalCapabilityReview (performance_technicalcapabilityreview):** `id` (UUID PK), `parameter_id` (FK TechnicalCapabilityParameter), `employee_id` (FK EmployeeProfile), `reviewer_id` (FK User), `score` (Decimal), `mentor_assessment` (TextField), `status` (CharField).

1.5 Goals & KRA Library (apps.goals)

- **KpiLibrary (goals_kpilibrary):** `id` (BigAuto PK), `title` (CharField), `position_id` (FK Position), `target_level_id` (FK JobLevel), `is_active` (Bool).
- **KpiLibraryDetail (goals_kpilibrarydetail):** `id` (BigAuto PK), `library_id` (FK KpiLibrary), `category_id` (FK KpiCategory), `goal_title` (CharField), `unit` (CharField), `target_value` (Decimal), `weight_percent` (Int), `is_compliance` (Bool).
- **GoalSet (goals_goalset):** `id` (BigAuto PK), `employee_id` (FK EmployeeProfile), `cycle_id` (FK PerformanceCycle), `assigned_by_id` (FK User), `status` (CharField), `version` (Int).
- **GoalItem (goals_goalitem):** `id` (BigAuto PK), `goal_set_id` (FK GoalSet), `category_id` (FK KpiCategory), `title` (CharField), `target_value` , `current_progress` (Decimal), `weight_percent` (Int).
- **KpiProgressEntry (goals_kpiprogressentry):** `id` (BigAuto PK), `goal_item_id` (FK GoalItem), `actual_value` , `progress_percent` (Decimal), `evidence_note` (TextField), `logged_by_id` (FK User).

- **KpiAuditTrail (goals_kpiaudittrail):** id (BigAuto PK), employee_id (FK EmployeeProfile), goal_set_id (FK GoalSet), action (CharField), change_reason (TextField), changed_by_id (FK User).

1.6 Feedback, Evidence, Intern & Audit (apps.feedback, intern, evidence, audit)

- **Feedback & Comments:** Continuous feedback entries with sender, recipient, type (PRAISE, IMPROVEMENT, CORRECTION), visibility (PUBLIC, MANAGER_ONLY, PEER), and threaded feedback comments.
- **EvidenceSubmission:** Milestone proof submissions with URL links, file attachments, and reviewer workflow (SUBMITTED, APPROVED, REVISION_REQUESTED).
- **InternTask:** Tasks assigned to mentees/interns with priority, deadlines, instructions, and mentor review verification.
- **InternForm & Submission:** Dynamic custom survey forms with multi-type questions (Rating, Text, Single-select) and structured JSON responses.
- **AuditLog (audit_auditlog):** Immutable compliance trail capturing actor, action, entity_type, entity_id, IP address, and JSON metadata.

2. Migrations Catalog

APPLICATION	MIGRATION IDENTIFIER	KEY OPERATIONS & SCHEMA ADDITIONS
accounts	0001_initial to 0005_seed_feedback_permissions	Custom User model, Email OTP, Password reset tokens, dynamic RolePermission schema, and default permission seeds.
organization	0001_initial to 0003_financialyear_...	Departments, Job Levels, Positions, Roles, Teams, Financial Years, Performance Categories, and Role-Level matrix.
employees	0001_initial to 0005_employeeprofile_...	EmployeeProfile linked to User and Department, joining dates, skill competencies JSON, and extended bio fields.
performance	0001_initial to 0006_technicalcapability...	Performance Cycles, Criteria rubrics, Appraisal workflows, Intern reply integration, Technical Capability parameters and review scoring.
goals	0001_initial to 0002_kpicategory_goalset_...	KRA & KPI Library catalog, GoalSet versioning, Goalltem weightings, KPI progress logging, and KPI audit trail.
feedback	0001_initial to 0002_feedback_is_anonymous_...	Feedback stream, threaded comments, anonymous feedback mode, and multi-tier visibility controls.

APPLICATION	MIGRATION IDENTIFIER	KEY OPERATIONS & SCHEMA ADDITIONS
evidence	0001_initial to 0002_initial	Milestone evidence uploads, file attachment paths, and manager verification status.
intern	0001_initial to 0002_interntask_requires_...	Intern task management, goal comments, self-appraisals, dynamic custom forms, and mentor review flags.
notifications	0001_initial to 0003_alter_notification_...	Scheduled announcement broadcaster, delivery time engines, and persona notification types.
attendance & training	0001_initial	Daily attendance records with check-in/out stamps, training courses catalog, and employee course enrollments.
audit	0001_initial	Immutable audit trail capturing actor, entity, action, and JSON metadata.

3. Technology Stack Breakdown

The platform is engineered using modern, type-safe, enterprise-grade frameworks across both backend and client tiers:

3.1 Client Tier (Frontend)

- **Core Framework:** React 19.2.5 with TypeScript ~6.0.2 for strict compile-time type validation.
- **Build Tooling:** Vite 8.0.9 with Rolldown-backed bundling delivering production builds in under 1.5s.
- **Styling:** Tailwind CSS v4.2.3 utilizing `@theme` CSS tokens, royal purple palette, and glassmorphic styling.
- **State Management:** Redux Toolkit 2.11.2 with RTK Query for normalized API caching, optimistic updates, and cache tags.
- **Routing:** React Router DOM v7.14.1 with multi-tier protected route wrappers and permission gates.
- **Data Visualization:** Recharts 3.8.1 for interactive performance pulse histograms, radar charts, and trend metrics.
- **Component Utilities:** Lucide React, React Toastify, React Signature Canvas.

3.2 Application Tier (Backend)

- **Framework:** Python 3.11 with Django 5.x delivering enterprise ORM, transactions, and security middleware.
- **REST Architecture:** Django REST Framework 3.15.0 with modular ViewSets, permission classes, and serializers.
- **Authentication:** SimpleJWT 5.3.0 supporting stateless Bearer tokens with sliding refresh windows.
- **API Documentation:** drf-spectacular 0.27.0 auto-generating OpenAPI 3.0 specification and interactive Swagger UI at `/api/docs/`.

- **Production Serving:** WhiteNoise resilient static delivery and Unicorn WSGI runtime.

3.3 Data & Infrastructure

- **Database:** PostgreSQL (Production) with ACID transactions and JSONField; SQLite for local testing.
- **Backend Hosting:** Render (`render.yaml`) automated container build and start commands.
- **Frontend Hosting:** Vercel (`vercel.json`) single-page application routing and global edge distribution.

4. Automated Test Suites & Verification

The backend includes comprehensive unit, integration, and security tests. All **25 automated tests pass with 0 errors** (execution time: 16.3s):

Module	Test Name	Validation Focus	Result
accounts	test_create_intern_user	User model creation with INTERN role and password hashing	PASSED
accounts	test_create_superuser	Superuser creation with admin flags and permissions	PASSED
accounts	test_duplicate_email_rejected	Unique email constraint enforcement	PASSED
accounts	test_roles_enumeration	All UserRole choice enumerations valid	PASSED
attendance	test_attendance_unique_constraint	Blocks duplicate daily attendance entries per employee	PASSED
attendance	test_training_course_and_enrollment	Course creation and employee completion calculation	PASSED
attendance	test_feedback_privacy_isolation	Validates feedback privacy scoping	PASSED
attendance	test_recognition_reward_creation	Peer recognition reward assignment	PASSED
attendance	test_audit_service_logging	Tamper-evident audit log creation	PASSED
employees	test_jwt_login_and_me_endpoint	JWT authentication and user profile retrieval	PASSED
employees	test_intern_least_privilege_scoping	Intern query isolation prevents accessing peer data	PASSED

MODULE	TEST NAME	VALIDATION FOCUS	RESULT
employees	test_manager_scoping	Manager visibility limited to direct team reports	PASSED
employees	test_hr_can_create_employee_atomically	Atomic creation of User + EmployeeProfile records	PASSED
employees	test_intern_cannot_create_employee	403 Forbidden on unauthorized employee creation	PASSED
goals	test_manager_can_create_goal_with_kpi	Manager goal assignment with associated KPI targets	PASSED
goals	test_intern_progress_logging	Progress update calculation and goal completion status	PASSED
goals	test_evidence_submission_and_review	Evidence submission and mentor review workflow	PASSED
performance	test_scoring_mathematical_precision	Weighted mathematical scoring engine validation	PASSED
performance	test_appraisal_lifecycle_and_visibility	Stage gate enforcement from draft to published	PASSED
reports	test_intern_can_view_own_summary	Personal performance summary report retrieval	PASSED
reports	test_intern_forbidden_other_report	Access denial for other employees' reports	PASSED
reports	test_manager_can_view_team_summary	Manager aggregated team metrics retrieval	PASSED
reports	test_intern_cannot_access_team_report	Blocks non-managers from team-level analytics	PASSED
reports	test_hr_can_view_organization_summary	Organization-wide analytics access for HR	PASSED
reports	test_export_performance_csv	CSV export generation and content verification	PASSED

5. Role-Based Access Control (RBAC) Matrix

Permissions are strictly segmented across 5 core roles following the Principle of Least Privilege (PoLP):

FUNCTIONAL AREA	CODE	SUPER ADMIN	HR PARTNER	TECH MANAGER	INTERN / EMP
Master Admin Access	ALL	Full Bypass	No Access	No Access	No Access
Role Management	ROLE_MANAGE	Edit & Assign	No Access	No Access	No Access
Permission Matrix	PERMISSION_MANAGE	Full Config	No Access	No Access	No Access
System Audit Logs	SYSTEM_AUDIT	View & Export	No Access	No Access	No Access
User / Employee Directory	USER_MANAGE	Full CRUD	Create & Edit	Team Read	Self Read
Org Structure & Teams	ORG_GOVERNANCE	Full CRUD	Manage Depts	Team Read	Read Only
Performance Cycles	CYCLE_MANAGE	Create / Launch	Manage / Close	Read Only	Active Only
Evaluation Rubrics	CRITERIA_MANAGE	Global Config	Form Builder	Read Only	No Access
Appraisal Final Publish	APPRAISAL_PUBLISH	Overrides	Final Release	No Access	No Access
Manager Evaluation	APPRAISAL_EVALUATE	Supervise	Calibrate	Score Reports	No Access
Self-Assessment	APPRAISAL_SELF	No Access	No Access	Self Only	Self Only
KRA / KPI Library	KPI_LIBRARY_MANAGE	Full Control	Manage Catalog	Assign Team	View Own
Goal Assignment	GOAL_ASSIGN	System-wide	Org-wide	Team Directs	No Access
Goal Progress Logging	GOAL_LOG	Full Access	Full Access	Team Reviews	Log Own

FUNCTIONAL AREA	CODE	SUPER ADMIN	HR PARTNER	TECH MANAGER	INTERN / EMP
Technical Capabilities	TECH_CAPABILITY	Manage Library	Overview	Review & Score	View Assigned
Evidence Submission	EVIDENCE_SUBMIT	No Access	No Access	No Access	Submit Proof
Evidence Verification	EVIDENCE_REVIEW	Oversee	View Only	Approve/Reject	No Access
Continuous Feedback	FEEDBACK_GIVE	Full Feed	Full Feed	Team & Peers	Give & Receive
Strategic Analytics	REPORT_VIEW_ALL	Strategic Pulse	Org Analytics	Team Pulse	Personal Only
Intern Tasks	TASK_MANAGE	View All	View All	Create / Assign	Complete Tasks
Announcement Broadcast	NOTIFICATION_POST	Broadcast Org	No Access	No Access	Read Inbox